Beam Current Limits

USPAS - Beam Physics with Intense Space Charge

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Outline I

1. Beam current limits in axisymmetric focusing channels

2. Beam current limits in quadrupolar focusing channels

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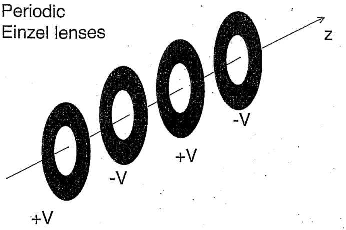
2. Beam current limits in quadrupolar focusing channels



PERIODIC SOLENOIDS

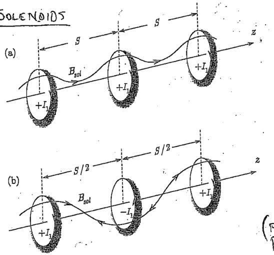


Figure 3.2. Schematic of magnet sets producing a periodic focusing solenoidal field with axial periodicity length S. In Fig. 3.2 (a), successive coils are spaced by S and have the same current polarity $+I_1, +I_1, \cdots$. In Fig. 3.2 (b), successive coils are spaced by $S/2$ and have alternating current polarities $+I_1, -I_1, +I_1, \cdots$.

(FIGURE FROM DAVIDSON & QIN, 2003) p.55 "PHYSICS OF INTENSE CHARGED PARTICLE BEAMS IN HIGH ENERGY ACCELERATORS"

1.1 Radial envelope equation

$$\tilde{r}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{r}' + \left(\frac{\gamma''}{2\gamma\beta^2}\right)\tilde{r} + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2\tilde{r} - \left(\frac{\langle p_\theta\rangle}{\gamma m\beta c}\right)^2\frac{1}{\tilde{r}^3} - \frac{\varepsilon_r^2}{\tilde{r}^3} - \frac{Q}{2\tilde{r}} = 0 \quad (1)$$

$$p_\theta \equiv \gamma m r^2\dot{\theta} + qrA_\theta, \quad (2)$$

$$\varepsilon_r \equiv \sqrt{\langle r^2\rangle\langle r'^2\rangle - \langle rr'\rangle^2 + \langle r^2\rangle\langle r^2\theta'^2\rangle - \langle r^2\theta'\rangle^2} \quad (3)$$

$$Q \equiv \frac{1}{mc^2\beta^2\gamma^3}\frac{q\langle\lambda(r)\rangle}{2\pi\epsilon_0} \quad (4)$$

1.2 Beam current limit in long solenoid

- Long solenoid: $\mathbf{B} = B_z \hat{z}$.
- No acceleration $\gamma' = \gamma'' = 0$.
- Set $p_\theta = 0$ and $\epsilon_r = 0$ to minimizing "repelling" forces.

$$\tilde{r}'' + \left(\frac{\omega_c}{2\gamma\beta c} \right)^2 \tilde{r} - \frac{Q}{2\tilde{r}} = 0 \tag{5}$$

Maximum beam perveance occurs at equilibrium, when $\tilde{r}' = \tilde{r}'' = 0$:

$$Q = \frac{1}{2} \left(\frac{\omega_c \tilde{r}}{\gamma\beta c} \right)^2 \tag{6}$$

1.3 Derivation from force balance at low energies

- Particle moves with angular velocity $\theta = \omega r$.
- Particle sees linear focusing force from solenoid field $B_z = \omega_c m/q$.
- Within beam envelope, particle sees linear defocusing force from space charge.

$$\underbrace{m\omega^2 r}_{\text{centrifugal}} + \underbrace{\frac{Qm\beta^2 c^2}{2\tilde{r}^2} r}_{\text{centrifugal}} = \underbrace{q v_\theta B_z}_{\text{magnetic}} \tag{7}$$

$$\omega^2 r + \frac{Q\beta^2 c^2}{2\tilde{r}^2} r = \omega_c \omega r \tag{8}$$

$$Q(\omega) = \frac{2\tilde{r}^2}{\beta^2 c^2} \left(\omega\omega_c - \omega^2\right) \tag{9}$$

$$\omega^* = \max_{\omega} Q(\omega) = \frac{\omega_c}{2} \tag{10}$$

$$Q(\omega^*) = \frac{1}{2} \left(\frac{\omega_c \tilde{r}}{\beta c}\right)^2 \tag{11}$$

Equivalent to result from envelope equations when $\gamma^2 \to 1$:

$$Q = \frac{1}{2} \left(\frac{\omega_c \tilde{r}}{\gamma\beta c}\right)^2 \tag{12}$$

1.4 Angular kick at solenoid entrance

Approximate solenoid field as step function:

$$B_z(z) = B_0\left[\Theta(z) + \Theta(L-z) - 1\right] = \begin{cases} 0 & \text{if } z < 0 \\ B_0 & \text{if } 0 \leq z \leq L \\ 0 & \text{if } z > L \end{cases} \tag{13}$$

$$\frac{\partial B_z}{\partial z} = B_0\left[\delta(z) - \delta(L-z)\right] \tag{14}$$

Use earlier result to relate longitudinal derivative to radial magnetic field:

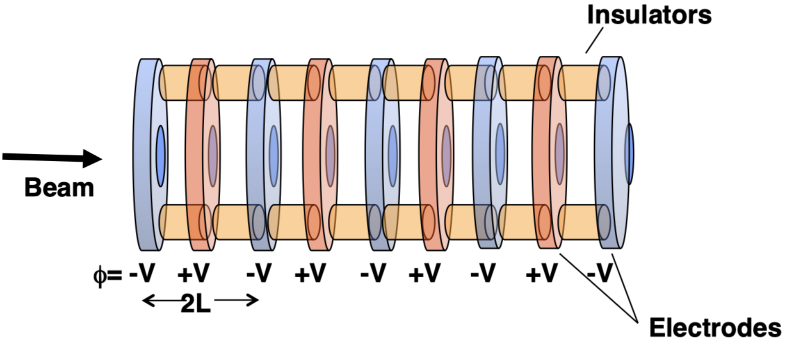
$$B_r(r,z) \approx -\frac{r}{2}\frac{\partial B_z}{\partial z} = \frac{rB_0}{2}\left[\delta(z) - \delta(L-z)\right] \tag{15}$$

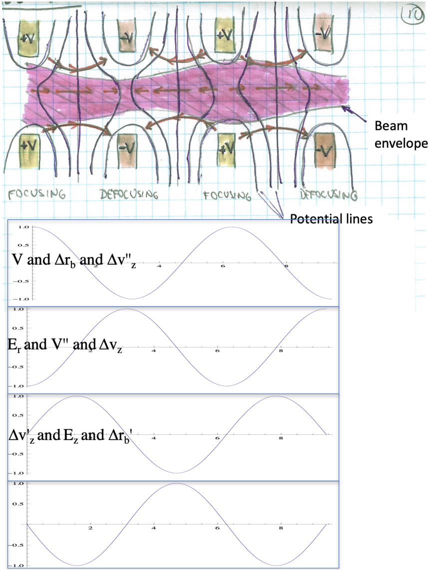
Compute change in θ component of momentum at $z = \epsilon$, just after solenoid entrance:

$$\begin{aligned}
\frac{d}{dt}p_\theta^*(r) &= qv_z B_r && (16)\\
\Delta p_\theta^*(r) &= q\int_{-\infty}^{\epsilon} B_r(r,z)dz && (17)\\
&= \frac{qrB_0}{2}\int_{-\infty}^{\epsilon}\left[\delta(z) - \delta(L-z)\right]dz && (18)\\
&= \frac{qrB_0}{2} && (19)
\end{aligned}$$

$$v_\theta = \frac{rqB_0}{2m} = \frac{r\omega_c}{2} \tag{20}$$

1.5 Beam current limits in periodic Einzel lens





$$V \quad \Rightarrow \quad v_z, \; V'' \quad \Rightarrow \quad E_r \quad \Rightarrow \quad \Delta r$$

Ignore centrifugal term, emittance term, and solenoid term in radial envelope equation.

$$\tilde{r}'' + \frac{\gamma'}{\gamma\beta^2}\tilde{r}' + \frac{\gamma''}{2\gamma\beta^2} - \frac{Q}{2\tilde{r}} = 0 \tag{21}$$

Assume non-relativistic beam: $\beta \ll 1$, $\gamma \approx 1$:

$$\gamma' = \beta\beta'\gamma^3 \approx \beta\beta' \tag{22}$$

$$\gamma'' \approx \beta'\beta' + \beta\beta'' \tag{23}$$

$$\tilde{r}'' + \frac{\beta'}{\beta}\tilde{r}' + \frac{1}{2}\left[\frac{\beta'^2}{\beta^2} + \frac{\beta''}{\beta}\right]\tilde{r} - \frac{Q}{2\tilde{r}} = 0 \tag{24}$$

Define new variable $R = \sqrt{\beta/\beta_0}\tilde{r}$, where β_0 is the initial velocity.

$$\tilde{r}' = \left(\frac{\beta}{\beta_0}\right)^{-1/2} R' - \frac{1}{2}\left(\frac{\beta}{\beta_0}\right)^{-3/2}\left(\frac{\beta'}{\beta}\right) R \tag{25}$$

$$\tilde{r}'' = \left(\frac{\beta}{\beta_0}\right)^{-1/2} R'' - \left(\frac{\beta}{\beta_0}\right)^{-3/2}\left(\frac{\beta'}{\beta_0}\right) R' + \frac{3}{4}\left(\frac{\beta}{\beta_0}\right)^{-5/2}\left(\frac{\beta'}{\beta_0}\right)^2 R - \frac{1}{2}\left(\frac{\beta}{\beta_0}\right)^{-3/2}\left(\frac{\beta}{\beta\right. \tag{26}$$

$$\boxed{R'' + \frac{3}{4}\left(\frac{\beta'}{\beta}\right)^2 R - \frac{1}{2}\frac{\beta}{\beta_0}\frac{Q}{R} = 0.} \tag{27}$$

Approximate potential as $\phi(r,z) = \phi_0 \cos(\pi z/L)$. From conservation of energy:

$$\Delta\left(\frac{1}{2}mv^2\right) = -\Delta\left(\frac{q\phi}{m}\right) \tag{28}$$

$$\frac{1}{2}m\left(v^2 - v_0^2\right) = -q\phi_0\cos\left(\frac{\pi z}{L}\right) \tag{29}$$

$$v^2 = v_0^2 - \frac{2q\phi_0}{m}\cos\left(\frac{\pi z}{L}\right) \tag{30}$$

Use this to rewrite $(\beta'/\beta)^2$:

$$v' = \frac{q\phi_0}{mv}\left(\frac{\pi}{L}\right)\sin\left(\frac{\pi z}{L}\right) \tag{31}$$

$$\frac{v'}{v} = \frac{q\phi_0}{mv^2}\left(\frac{\pi}{L}\right)\sin\left(\frac{\pi z}{L}\right) \tag{32}$$

$$\left(\frac{\beta'}{\beta}\right)^2 = \left(\frac{q\phi_0}{mv}\right)^2\left(\frac{\pi}{L}\right)^2\sin^2\left(\frac{\pi z}{L}\right) \tag{33}$$

Assume $v_0^2 \gg \frac{2q\phi_0}{m}$, so that $v \approx v_0$.

$$\left(\frac{\beta'}{\beta}\right)^2 \approx \left(\frac{q\phi_0}{mv_0}\right)^2 \left(\frac{\pi}{L}\right)^2 \sin^2\left(\frac{\pi z}{L}\right) \tag{34}$$

For equilibrium ($R'' = R' = 0$), look at averages over z:

$$\overline{\left(\frac{\beta'}{\beta}\right)^2} = \frac{1}{2}\left(\frac{q\phi_0}{mv_0}\right)^2 \left(\frac{\pi}{L}\right)^2 \tag{35}$$

$$\overline{\left(\frac{\beta}{\beta_0}\right)} = 0 \tag{36}$$

$$\overline{R} = \overline{\left(\frac{\beta}{\beta_0}\right)^{1/2}}\tilde{r} = \tilde{r} \tag{37}$$

Inserting these terms in the envelope equation with $R'' = 0$ gives the maximum beam perveance:

$$\boxed{Q_{max} = \frac{3\pi^2}{4}\left(\frac{q\phi_0}{m\beta_0^2 c^2}\right)^2 \left(\frac{\tilde{r}}{L}\right)^2} \tag{38}$$

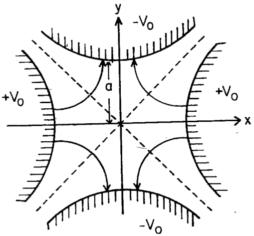
Outline

1. Beam current limits in axisymmetric focusing channels

2. Beam current limits in quadrupolar focusing channels

FROM REISER, p.112

$E_x = -E'x$
$E_y = E'y$



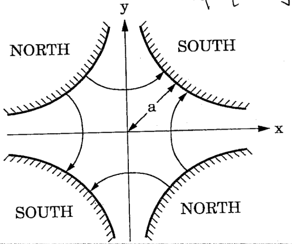
$F_x = -qE'x$
$F_y = qE'y$

ELECTROSTATIC QUADS

Figure 3.15. Electrodes and force lines in an electrostatic quadrupole.

$B_x = B'y$
$B_y = B'x$

$F_x = -q v_z B' x$
$F_y = q v_z B' y$



MAGNETIC QUADS

2.1 RMS envelope equations

$$\tilde{x}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{x}' + \kappa_x(s)\tilde{x} - \frac{\varepsilon_x^2}{\tilde{x}^3} - \frac{Q}{2\left(\tilde{x}+\tilde{y}\right)} = 0 \tag{39}$$

$$\tilde{y}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{y}' + \kappa_y(s)\tilde{y} - \frac{\varepsilon_y^2}{\tilde{y}^3} - \frac{Q}{2\left(\tilde{x}+\tilde{y}\right)} = 0 \tag{40}$$

$$Q = \frac{q\lambda}{2\pi\epsilon_0 mc^2\beta^2\gamma^3} \tag{41}$$

$$\varepsilon_x(s) = \sqrt{\langle xx\rangle\langle x'x'\rangle - \langle xx'\rangle\langle xx'\rangle} \tag{42}$$

$$\varepsilon_y(s) = \sqrt{\langle yy\rangle\langle y'y'\rangle - \langle yy'\rangle\langle yy'\rangle} \tag{43}$$

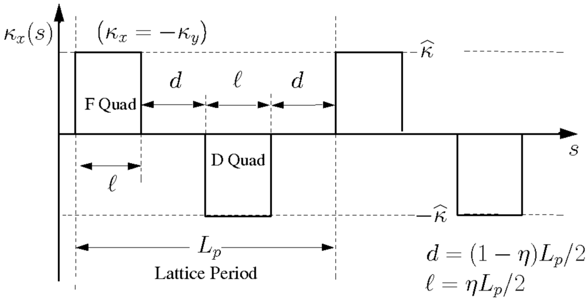
$$\kappa_x(s) = +\frac{qG}{\gamma\beta mc} \tag{44}$$

$$\kappa_y(s) = -\frac{qG}{\gamma\beta mc} \tag{45}$$

Let $\kappa_x(s) = f(s)\kappa$ and $\kappa_y(s) = -\kappa_x(s)$. Ignoring acceleration ($\gamma' = 0$):

$$\tilde{x}'' + \kappa f(s)\tilde{x} - \frac{\varepsilon_x^2}{\tilde{x}^3} - \frac{Q}{2\left(\tilde{x} + \tilde{y}\right)} = 0 \tag{46}$$

$$\tilde{y}'' - \kappa f(s)\tilde{y} - \frac{\varepsilon_y^2}{\tilde{y}^3} - \frac{Q}{2\left(\tilde{x} + \tilde{y}\right)} = 0 \tag{47}$$



$\eta =$ Occupancy $\in (0, 1]$

$$f(s) = \begin{cases} +1 & \text{if } 0 \leq s < L\eta/2 \\ -1 & \text{if } L - \eta L/2 \leq s < L + L\eta/2 \\ +1 & \text{if } 2L - L\eta/2 \leq s < 2L \end{cases} \tag{48}$$

To derive the maximum beam current, approximate $f(s)$, $\tilde{x}(s)$, and $\tilde{y}(s)$ as sinusoidal functions at the lattice frequency. Then plug into envelope equation, ignore fast-oscillating terms, and solve for equilibrium.

Start with the focusing strength $f(s)$:

$$f(s) = \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi s}{L}\right) \tag{49}$$

$$a_n = \frac{1}{L}\int_0^{2L} f(s) \cos\left(\frac{n\pi s}{L}\right) ds \tag{50}$$

$$\boxed{f(s) \approx a_1 \cos\left(\frac{\pi s}{L}\right) = \frac{4}{\pi} \sin\left(\frac{\pi \eta}{2}\right) \cos\left(\frac{\pi s}{L}\right)} \tag{51}$$

Approximate rms envelope evolution as:

$$\tilde{x}(s) = \frac{\tilde{r}_0}{\sqrt{2}} \left[1 + \delta \cos\left(\frac{\pi s}{L}\right)\right] \tag{52}$$

$$\tilde{y}(s) = \frac{\tilde{r}_0}{\sqrt{2}} \left[1 - \delta \cos\left(\frac{\pi s}{L}\right)\right] \tag{53}$$

Now evaluate terms in rms envelope equations: $\tilde{x}''$, $f(s)\tilde{x}$, etc.

$$\tilde{x}'' = -\frac{\tilde{r}_0}{\sqrt{2}} \delta \frac{\pi^2}{L^2} \cos\left(\frac{\pi s}{L}\right) \tag{54}$$

$$\tilde{y}'' = +\frac{\tilde{r}_0}{\sqrt{2}} \delta \frac{\pi^2}{L^2} \cos\left(\frac{\pi s}{L}\right) \tag{55}$$

$$f(s)\tilde{x}(s) = \frac{4\tilde{r}_0}{\sqrt{2}\pi} \sin\left(\frac{\pi\eta}{2}\right) \cos\left(\frac{\pi s}{L}\right) \left[1 + \delta \cos\left(\frac{\pi s}{L}\right)\right], \tag{56}$$

$$= \frac{4\tilde{r}_0}{\sqrt{2}\pi} \sin\left(\frac{\pi\eta}{2}\right) \left[\cos\left(\frac{\pi s}{L}\right) + \delta \cos^2\left(\frac{\pi s}{L}\right)\right]. \tag{57}$$

Ignore fast oscillating component of $\cos^2(\pi s/L)$:

$$\cos^2\left(\frac{\pi s}{L}\right) = \frac{1}{2} + \frac{1}{2}\cos\left(\frac{2\pi s}{L}\right) \approx \frac{1}{2}. \tag{58}$$

This gives:

$$f(s)\tilde{x}(s) \approx \frac{4\tilde{r}_0}{\sqrt{2}\pi} \sin\left(\frac{\pi\eta}{2}\right) \left[\cos\left(\frac{\pi s}{L}\right) + \frac{\delta}{2}\right], \tag{59}$$

$$f(s)\tilde{y}(s) \approx \frac{4\tilde{r}_0}{\sqrt{2}\pi} \sin\left(\frac{\pi\eta}{2}\right) \left[\cos\left(\frac{\pi s}{L}\right) - \frac{\delta}{2}\right]. \tag{60}$$

Plugging these appoximations into rms envelope equations:

$$-\left[\delta\frac{\pi^2}{L^2}-\frac{4\kappa}{\pi}\sin\left(\frac{\eta\pi}{2}\right)\right]\cos\left(\frac{\pi s}{L}\right)+\frac{2\kappa\delta}{\pi}\sin\left(\frac{\eta\pi}{2}\right)=\frac{Q}{2\tilde{r}_0^2}, \tag{61}$$

$$+\left[\delta\frac{\pi^2}{L^2}-\frac{4\kappa}{\pi}\sin\left(\frac{\eta\pi}{2}\right)\right]\cos\left(\frac{\pi s}{L}\right)+\frac{2\kappa\delta}{\pi}\sin\left(\frac{\eta\pi}{2}\right)=\frac{Q}{2\tilde{r}_0^2}. \tag{62}$$

Solve for δ and Q:

$$\delta=\frac{4\kappa L^2}{\pi^3}\sin\left(\frac{\eta\pi}{2}\right) \tag{63}$$

$$Q=\frac{4\kappa\delta}{\pi}\sin\left(\frac{\eta\pi}{2}\right)\tilde{r}_0^2 \tag{64}$$

$$\boxed{Q_{max}\approx\left(\frac{2\kappa L\eta}{\pi}\right)^2\left(\frac{\sin(\eta\pi/2)}{(\eta\pi/2)}\right)^2\tilde{r}_0^2} \tag{65}$$

2.2 Continuous focusing

$$\tilde{x}'' + \omega_0^2 \tilde{x} - \frac{\varepsilon^2}{\tilde{x}^3} - \frac{Q}{2\left(\tilde{x}+\tilde{y}\right)} = 0, \tag{66}$$

$$\tilde{y}'' + \omega_0^2 \tilde{y} - \frac{\varepsilon_y^2}{\tilde{y}^3} - \frac{Q}{2\left(\tilde{x}+\tilde{y}\right)} = 0. \tag{67}$$

At equilibrium ($\tilde{x}'' = \tilde{y}'' = 0$), the defocusing space charge term balances the external focusing term. This occurs at the equilibrium rms radius $\tilde{r}_0 = \sqrt{2}\tilde{x} = \sqrt{2}\tilde{y}$.

$$Q_{max} = 2\omega_0^2 \tilde{r}_0^2 \tag{68}$$

Comparing to Eq. (65) gives the oscillation frequency ω_0^2 for the smooth-focusing approximation of the FODO channel:

$$\omega_0^2 = \frac{2\kappa^2 L^2 \eta^2}{\pi^2} \left(\frac{\sin(\eta\pi/2)}{(\eta\pi/2)}\right)^2 \tag{69}$$

Define the phase advance $\sigma_0 = \omega_0/(2L)$ to eliminate L:

$$\omega_0^2 = \frac{\eta k \sigma_0}{\sqrt{2}\pi} \frac{\sin(\eta\pi/2)}{(\eta\pi/2)} \tag{70}$$

$$Q_{max} = 2\omega_0^2 \tilde{r}_0^2 = \frac{\sqrt{2}\eta k \sigma_0}{\pi} \frac{\sin(\eta\pi/2)}{(\eta\pi/2)} \tilde{r}_0^2. \tag{71}$$

Define $r_b \equiv \sqrt{2}\tilde{r}_0$ as beam edge radius and r_p as pipe radius.

$$Q_{max} = \frac{\eta k \sigma_0}{\sqrt{2}\pi} \frac{\sin(\eta\pi/2)}{(\eta\pi/2)} r_b^2. \tag{72}$$

Magnetic quadrupoles:

$$k = \frac{q}{\gamma m \beta c} \frac{dB}{dx} \sim \frac{q}{\gamma m \beta c} \frac{B}{r_p}. \tag{73}$$

Electric quadrupoles:

$$k = \frac{q}{\gamma m \beta^2 c^2} \frac{dE}{dx} \sim \frac{2qV_q}{\gamma m \beta^2 c^2}, \tag{74}$$

where $V_q = \frac{1}{2} \frac{dE}{dx} r_p^2$.

2.3 Summary

EINZEL LENS	SOLENOIDS	QUADRUPOLE FOCUSING
$Q_{max} \cong \frac{3\pi^2}{8}\left(\frac{q\Phi_0}{m v_0^2}\right)^2\left(\frac{r_b}{L}\right)^2$	$Q_{max} = \left(\frac{\omega_c r_b}{2\gamma\beta c}\right)^2$	$Q_{max} \cong \frac{\eta\sigma_0}{\sqrt{2\pi}}\left(\frac{\sin\frac{\eta\pi}{2}}{\frac{\eta\pi}{2}}\right)\begin{bmatrix} \frac{B r_b}{[B\rho]}\left[\frac{r_b}{r_p}\right] \\ \frac{2qV_q}{\gamma m v_z^2}\left[\frac{r_b^2}{r_p^2}\right] \end{bmatrix}$ MAGNETIC / Electric

For low-energy beams ($\gamma \approx 1$), $\lambda_{max} \sim Q_{max}V$ where $V = \frac{1}{2}m\beta^2c^2/q$.

$\lambda_{max} \propto \frac{\Phi_0^2}{V}$	$\lambda_{max} \propto \frac{q}{m}B^2 r_p^2$	$\lambda_{max} \propto \begin{cases} B_1 V^{1/2} r_p \\ V_q \end{cases}$

NOTE
Φ_0 = Voltage between Einzel lenses
V_q = Voltage on a quad relative to ground
V = particle energy / q